



# **Crypto - Risk Lens**

**Aditya Kate – 23107126**

**Suman Manik – 23107056**

**Tanmay Harmalkar – 23107099**

**Project Guide**

**Ms. Rajashri Chaudhari**

# **Sustainable Development Goals (SDG) mapped**

- **SDG 9: Industry, Innovation and Infrastructure**

A technical Infrastructure for real-time risk monitoring in Crypto Markets. Building resilient infrastructure and sustainable industrialization and foster innovation.

- **SDG 8: Decent Work and Economic Growth**

A safer, more informed investment decisions, which helps avoid destructive speculation and bubbles, contributes to stable, sustainable speculation activity in digital assets.

# Outline

- SDG Mapped
- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

# Introduction

- Crypto-Risk Lens analyzes cryptocurrency markets by combining price, volume, news, and social media sentiment.
- Using Machine Learning and NLP, it detects market trends, measures volatility and risk, and extracts sentiment signals from crypto news and discussions.
- The goal is to understand how market behavior and public sentiment together affect crypto market stability and risk.

## **Motivation:**

- Crypto markets are highly volatile and strongly affected by news and public sentiment, but most tools focus only on price charts and technical indicators.
- An AI-based system combining market data and sentiment analysis is needed for better risk understanding and decision making.

# Objectives

- To collect and preprocess real-time cryptocurrency market data and news data and convert them into structured market and sentiment features.
- To extract sentiment trends and dominant themes from crypto-related news using natural language processing techniques.
- To identify market regimes and risk levels of crypto assets using unsupervised and statistical learning methods.
- To present clear and understandable risk and sentiment insights to help users to make informed decisions without giving direct buy or sell advice.

# Literature Survey of the existing system

| Sr. No. | Title                                                                                                    | Author(s)                                                                           | Year | Methodology                                                                                                                       | Drawbacks                                                      |
|---------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 1.      | [1] Bitcoin Price Prediction Based on Sentiment Analysis                                                 | Jutur Manogna, Gogineni Sravan Chowdary, Gogineni Meghana, Priyanka C Nair          | 2024 | Uses Twitter feelings and Bitcoin prices from apps to guess short-term prices. Simple sentiment tools help predict ups and downs. | Only good for short time; bad Twitter data can mess it up.     |
| 2.      | [2] Cryptocurrency Insight: A Multiplatform Sentiment Analysis Framework for Price Prediction            | Andrew Coutinho, Reshu Agarwal                                                      | 2024 | Checks feelings from many sites and uses basic ML to guess crypto prices.                                                         | Hard to mix data from different places without errors.         |
| 3.      | [3] Exploring the Effectiveness of Adding Sentiment Analysis into Random Forest to Predict Bitcoin Price | Vincent Oei, Pieter Effendy, Lili Ayu Wulandhari, Islam Nur Alam                    | 2024 | Random Forest tree model plus social media feelings to predict Bitcoin.                                                           | Fits too well to one type of data; doesn't work for all coins. |
| 4.      | [4] Risk Analysis of Stablecoins: Exploring Blockchain Integration and Market Impact                     | Shalini Singh, Shubhanker Yadav, Anshupriya Tirkey, Beronica Kullu, Kartikeya Singh | 2024 | Looks at old market data for stablecoins and blockchain risks.                                                                    | Uses past info only; no live checks.                           |

# Literature Survey of the existing system

| Sr. No. | Title                                                                                                                               | Author(s)                                                                                                            | Year | Methodology                                                                | Drawbacks                                                        |
|---------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------|------------------------------------------------------------------|
| 5.      | [5] Cryptocurrency Price Prediction Using LSTM and FEDformer Enhanced by Sentiment Analysis                                         | Sumin Li, Rentao Wang, Yudong Wan, Jincheng Hu                                                                       | 2025 | Mixes LSTM for quick changes and FEDformer for long trends, plus feelings. | Needs lots of computer power; feelings part can be weak.         |
| 6.      | [6] HSIF: A Transformer-Based Cross-Attention Framework for Cryptocurrency Trend Forecasting via Multimodal Sentiment–Market Fusion | Saeed Mohammadi Dashtaki, Mehdi Hosseini Chagahi, Aein Bahadori, Behzad Moshiri, Md. Jalil Piran, Allahyar Montazeri | 2025 | Smart Transformer model mixes market data and feelings from text.          | Uses too much power; needs perfect feeling data.                 |
| 7.      | [7] Leveraging Machine Learning for Anomaly Detection Methods in Cryptocurrency: A Data-Driven Study                                | Achraf Yahia, Yassine Mouhssine, Abdelkader El Alaoui, Said Ouatik Elalaoui                                          | 2024 | Simple ML tools like Isolation Forest spot weird price jumps.              | No labels means it misses some issues; works different per coin. |
| 8.      | [8] Machine Learning in DeFi: Credit Risk Assessment and Liquidation Prediction                                                     | Georgios Palaiokrassas, Sandro Scherrers, Eftychia Makri, Leandros Tassioulas                                        | 2024 | ML checks loan risks and predicts when loans fail in DeFi.                 | DeFi data is messy; hard to explain why it decides.              |

# Limitations of existing systems

1. **Survey Analysis:** Current solutions focus mainly on price prediction or sentiment analysis separately; lack integrated real-time fusion of market data, news sentiment, volatility, and regime detection in a single system.
2. **Problem Status:** Awareness about crypto volatility and sentiment impact exists, but practical, real-time risk monitoring platforms combining prediction, sentiment, and visualization are limited.
3. **Observations & Improvements:** ML and deep learning models show strong prediction accuracy; sentiment improves performance. Improvements needed in real-time sentiment extraction, regime classification, explainable risk scoring, and integrated dashboards.
4. **Technology :** Transformer and LSTM based Time Series perform well for forecasting; limitations include high computation cost, overfitting in volatile markets.

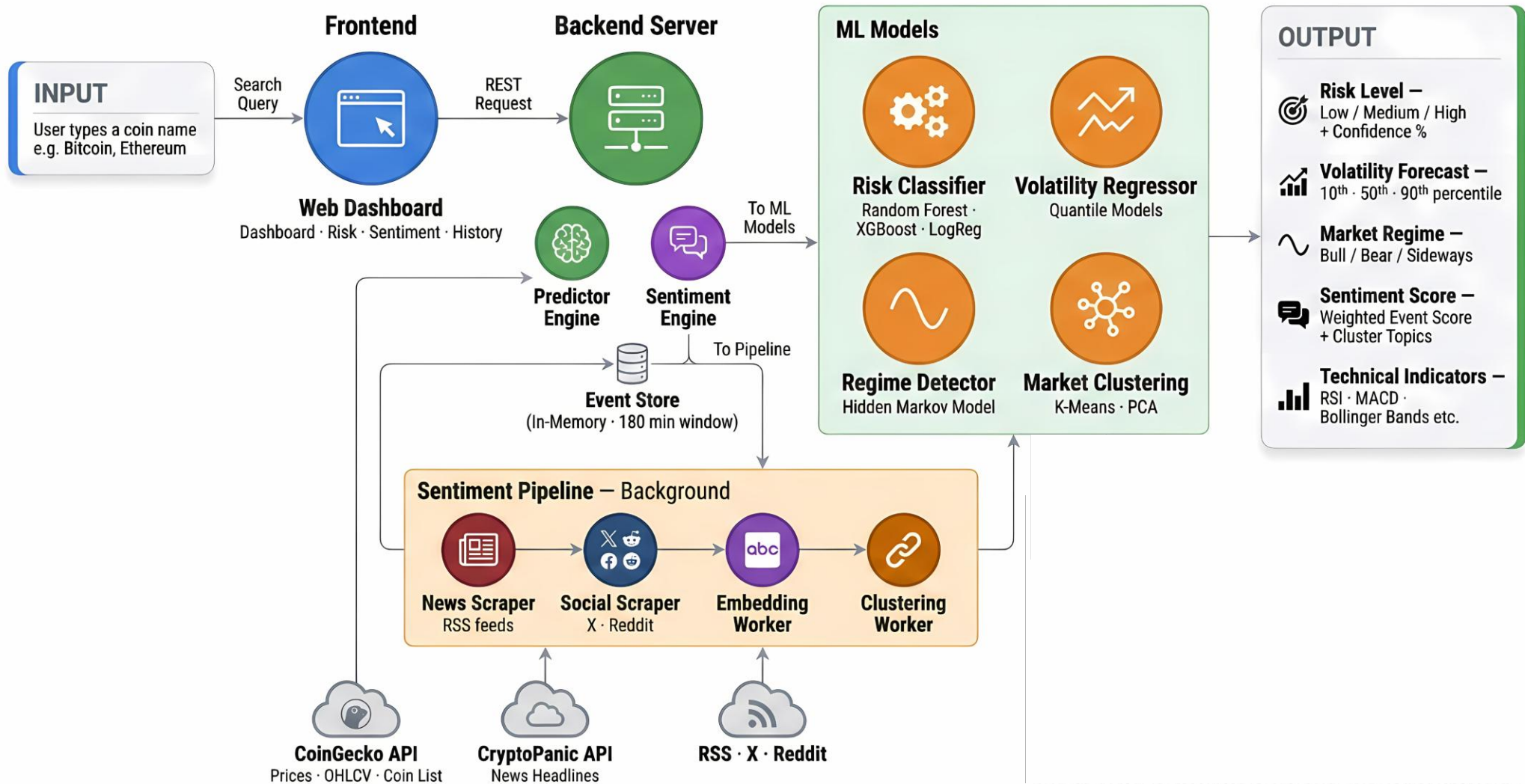
# Problem statement

Cryptocurrency markets change very quickly due to price movements, trading behavior, and news sentiment. Most existing tools only analyze historical prices and technical indicators.

## Problem Elaboration :

1. **Lack of integrated analysis:** Price data and sentiment signals are analyzed separately, no unified risk framework combines both effectively.
2. **Limited real-time sentiment tracking:** Most systems do not continuously extract and quantify sentiment from live crypto news and discussions.
3. **Absence of regime detection:** Existing approaches fail to identify bull, bear, or high-volatility market phases using AI-based models.
4. **Weak risk prediction support:** Investors lack predictive intelligence that connects market behavior and sentiment to long-term stability and risk assessment.

# System Design – Architecture Diagram



# Implementation – Landing Page

 CryptoRisk Lens

Technology

Launch System

• INTELLIGENT RISK ANALYSIS

## Navigate the market with **absolute clarity.**

Stop guessing. We transform complex volatility, technicals, and sentiment into beautifully simple risk metrics you can trust.

Enter Dashboard >

View Market Sentiment 

# Implementation – Landing Page

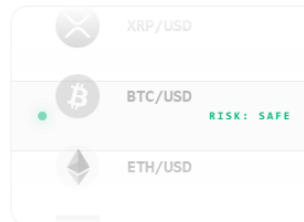
**Powerful analysis.**  
**Simple design.**

Real market intelligence shouldn't be complicated. We process millions of data points, so you see precisely what matters.



## Live Risk Categorization

Instantly know if it's safe to enter. Our system continuously analyzes 30+ technical metrics to provide clear risk bounds.



ms

Lightning-fast, real-time market data processing.



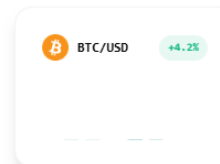
## Trend Detection

Automatically identifies market momentum—whether bull, bear, or completely sideways.



## Smarter Indicators

RSI, MACD, and Bollinger Bands tracked natively and continuously, displayed only when there's an actionable signal.



# Implementation – Landing Page

SYSTEM FLOW

## Data doesn't sleep.

From raw market feeds to clear dashboard visuals in milliseconds. Constant updates. Constant edge.



**Ingest**  
Live ticks



**Extract**  
Processing



**Score**  
Analysis



**Render**  
Dashboards

# **Implementation – Dashboard**



Market Overview

Real-time analysis for Ethereum

Search cryptocurrency...

Live

Overview

Sentiment

PRICE ANALYTICS

Ethereum Price



\$2,125.06

+1.94%

24h Volume



29.09B

11.34% of cap

Market Cap



256.46B

Rank #2

Volatility (7d)



2.16%

Low

Ethereum/USD Price Chart

7-day historical price movement

Line

Candlestick

Live



RISK ANALYTICS

Current Risk Level



22/100

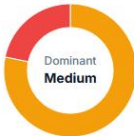
Stable

Low volatility, strong liquidity, neutral sentiment.

Model Confidence

78%

Risk Distribution



Low Risk

0.0%

Medium Risk

78.2%

High Risk

21.7%

Technical Indicators

TA-Lib powered analysis

Momentum

RSI (14)

62.38

Stochastic RSI

28.04

MACD

41.67

MACD Signal

29.49

MACD Histogram

12.18

Momentum

149.46

ROC

7.52

Volume

OBV

-116.368

MFI

43.34

Volume SMA Ratio

1.00

Trend

ADX

21.38

Aroon Oscillator

50.00

CCI

106.02

TRIX

0.2058

Oscillators

Williams %R

-21.43

Ultimate Oscillator

57.43

Balance of Power

0.5745

Volatility

ATR (14)

50.94

BB Width

0.1421

Volatility (7d)

2.31%

Volatility (30d)

1.89%

Price Action

Drawdown

-6.96%

Max Drawdown (30d)

-17.51%

Price/SMA50 Ratio

1.0677

Returns (1d)

1.73%

Market Status

Live Feed

# **Implementation – Sentiment**

## Real-Time Sentiment

Live news, social media, and forum posts for **Ethereum**

Generate Summary

Live

### SENTIMENT METRICS

Event Score

34.65

Total Mentions

125

Authoritative

13%

Social Signals

31%

### Live Feed

100 articles



NEWSBTC RSS Authoritative

5h ago

#### Manufacturing The Bitcoin Reserve: Inside The Trump Family's 11,000-Miner Expansion At American Bitcoin

Bitcoin is regaining momentum after reclaiming the \$70,000 level, signaling renewed strength following weeks of consolidation and volatile price action. The move above this key psychological threshold has helped stabilize sentiment across the market, as investors assess whether the recent correction has begun to transition into a new accumulation phase. Related Reading: The Quiet...



NEWSBTC RSS Authoritative

5h ago

#### Solana (SOL) Rally Builds, Traders Watch Critical \$100 Test

Solana started a fresh increase above the \$88 zone. SOL price is now consolidating above \$90 and might aim for more gains above the \$95 zone. SOL price started a fresh upward move above the \$85 and \$88 levels against the US Dollar. The price is now trading above \$90 and the 100-hourly simple moving average. There is a bullish trend line forming with support at \$89 on the hourly ch...



NEWSBTC RSS Authoritative

5h ago

#### US Should Act On Bitcoin, Not Just Praise It, Ex-Advisor To Trump Says

The US government is sitting on roughly 378,372 Bitcoin worth more than \$24 billion, according to data from Arkham Research. Yet more than a year after US President Donald Trump signed an executive order establishing a Strategic Bitcoin Reserve, no new Bitcoin has been purchased. Related Reading: Iran's Crypto Market Shaken As Outflows Skyrocket 700% The government has not...



NEWSBTC RSS Authoritative

5h ago

#### Bitcoin Historically Bottoms Between These MVRV Levels—Where Are They Now?

An analyst has highlighted how Bitcoin has consistently bottomed out between the 1.0 and 0.8 MVRV pricing bands during the past decade. Bitcoin Still Hasn't Breached Below 1.0 MVRV Band In This Cycle In a new post on X, analyst Ali Martinez has talked about historical Bitcoin bottoms from the perspective of the MVRV Pricing Bands. This is an on-chain pricing model for BTC...

Market Status

Live Feed

# Technologies and methodologies

## Frontend:

- **React + TypeScript + Vite** : Builds dynamic, component-based dashboard UI.
- **Recharts + D3** : Visualizes price trends, volatility, sentiment scores, and risk indicators.
- **Native Fetch with custom API Client**: Connects frontend with backend APIs for real-time data updates.

## Backend:

- **FastAPI/Flask**: Handles API requests, serves predictions, and manages real-time market data processing.
- **Pandas/NumPy**: Used for data cleaning, preprocessing, feature engineering, and numerical computations.
- **Scikit-learn**: Implements machine learning models for market regime detection and risk prediction.
- **TA-Lib**: Generates technical indicators (RSI, MACD, EMA, Bollinger Bands, etc.) for market behaviour analysis.

# Technologies and methodologies

## Methodology:

- Data collection: Gather historical OHLCV and volume from CoinGecko plus real-time crypto news and social data.
- Preprocessing & features: Clean/normalize data and compute technical indicators (RSI, MACD, volatility, drawdown, etc.).
- Modeling: Train ML models for risk classification, volatility forecasting, clustering, and HMM-based regime detection; use NLP embeddings to extract sentiment/event signals.
- Frontend: Use React + Recharts to visualize prices, risk, volatility, regimes, and sentiment in a real-time dashboard.

# Technologies and methodologies

## Algorithms

- **Regression** – Linear Regression, Quantile Regression: Estimate how market indicators influence crypto risk and volatility.
- **Classification** – Logistic Regression, Random Forest, XGBoost: Categorize the market into low, medium, or high-risk states.
- **Clustering** – K-Means, Hierarchical Clustering: Identify recurring market behaviour patterns without labelled data.
- **Time-Series / Regime Detection** – Hidden Markov Model, Change-Point Detection: Detect hidden market regimes and sudden structural shifts.
- **Dimensionality Reduction** – PCA: Reduce correlated technical and sentiment indicators into key components for analysis.

# Technologies and methodologies

## Datasets

- **Cryptocurrency historical market data:** Directly fetched from CoinGecko (prices and volume for major coins), then converted to OHLCV and technical features.
- **Real-time crypto news and discussions:** Live RSS feeds from major crypto news sites and official blogs, plus Reddit, X (formerly Twitter) (via Nitter), and BitcoinTalk posts, scraped and processed into sentiment/event signals.

# Conclusion

The Crypto-Risk Lens, an AI-based market and sentiment risk analysis system that delivers near real-time crypto risk assessment by integrating market data, technical indicators, and news/social sentiment. By combining machine learning, statistical modeling, and predictive analytics, the system supports investors, analysts, and researchers in making more informed decisions and deepens understanding of volatility, market regimes, and overall risk in cryptocurrency markets.

# References

- [1] Jutur Manogna, Gogineni Sravan Chowdary, Gogineni Meghana, & Priyanka C Nair. (2024). "Bitcoin Price Prediction Based on Sentiment Analysis." IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10440877/>. [Accessed: Feb. 22, 2026].
- [2] Andrew Coutinho, & Reshu Agarwal. (2024). "Cryptocurrency Insight: A Multiplatform Sentiment Analysis Framework for Price Prediction." IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/11013419/>. [Accessed: Feb. 22, 2026].
- [3] Vincent Oei, Pieter Effendy, Lili Ayu Wulandhari, & Islam Nur Alam. (2024). "Exploring the Effectiveness of Adding Sentiment Analysis into Random Forest to Predict Bitcoin Price." IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10903842/>. [Accessed: Feb. 22, 2026].
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- [5] Sumin Li, Rentao Wang, Yudong Wan, & Jincheng Hu. (2025). "Cryptocurrency Price Prediction Using LSTM and FEDformer Enhanced by Sentiment Analysis." IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/11299632/>. [Accessed: Feb. 22, 2026].

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- [6] Saeed Mohammadi Dashtaki, Mehdi Hosseini Chagahi, Aein Bahadori, Behzad Moshiri, Md. Jalil Piran, & Allahyar Montazeri. (2025). "HSIF: A Transformer-Based Cross-Attention Framework for Cryptocurrency Trend Forecasting via Multimodal Sentiment–Market Fusion." IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10820123>. [Accessed: Feb. 22, 2026].
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**Thank You...!!**